Recap: The consumer problems

Utility Maximization Problem

$$\max_{x \in \mathbb{R}^n_+} u(x) \text{ such that } p \cdot x \leq w.$$

- Choice correspondence: Marshallian demand $x(p, w)$
- Value function: indirect utility function $v(p, w)$

Expenditure Minimization Problem

$$\min_{x \in \mathbb{R}^n_+} p \cdot x \text{ such that } u(x) \geq \bar{u}.$$

- Choice correspondence: Hicksian demand $h(p, \bar{u})$
- Value function: expenditure function $e(p, \bar{u})$

Key questions addressed by consumer theory

Already addressed

- What problems do consumers solve?
- What do we know about the solutions to these CPs generally? What about if we apply restrictions to preferences?
- How do we actually solve these CPs?
- How do the value functions and choice correspondences relate within/across UMP and EMP?

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- How do we actually solve these CPs?
- How do the value functions and choice correspondences relate within/across UMP and EMP?

Still to come

- How do we measure consumer welfare?
- How should we calculate price indices?
- When and how can we aggregate across heterogeneous consumers?

Outline

- The welfare impact of price changes
- Price indices
  - Price indices for all goods
  - Price indices for a subset of goods
- Aggregating across consumers
- Optimal taxation

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Quantifying consumer welfare I

Key question

How much better or worse off is a consumer as a result of a price change from p to p'?

Applies broadly:

- Actual price changes
- Taxes or subsidies
- Introduction of new goods

Quantifying consumer welfare II

Challenge will be to measure how "well off" a consumer is without using utils—recall preference representation is ordinal
This rules out a first attempt:

$$\Delta u = v(p', w) - v(p, w)$$

To get a dollar-denominated measure, we can ask one of two questions:

1. How much would consumer be willing to pay for the price change?
   Fee + Price change $\sim$ Status quo
2. How much would we have to pay consumer to miss out on price change?
   Price change $\sim$ Status quo + Bonus

Quantifying consumer welfare III

Both questions fundamentally ask "how much money is required to achieve a fixed level of utility before and after the price change?"

$$\text{Variation} = e(p, u_{\text{reference}}) - e(p', u_{\text{reference}})$$

For our two questions,

1. How much would consumer be willing to pay for the price change?
   Reference: Old utility ($u_{\text{reference}} = \bar{u} \equiv v(p, w)$)
2. How much would we have to pay consumer to miss out on price change?
   Reference: New utility ($u_{\text{reference}} = \bar{u}' \equiv v(p', w)$)

Compensating and equivalent variation

Definition (Compensating variation)

The amount less wealth (i.e., the fee) a consumer needs to achieve the same maximum utility at new prices (p') as she had before the price change (at prices p):

$$\text{CV} \equiv e\big(p, v(p,w)\big) - e\big(p', v(p,w)\big) = w - e\big(p', \underbrace{v(p,w)}_{\equiv \bar{u}}\big).$$

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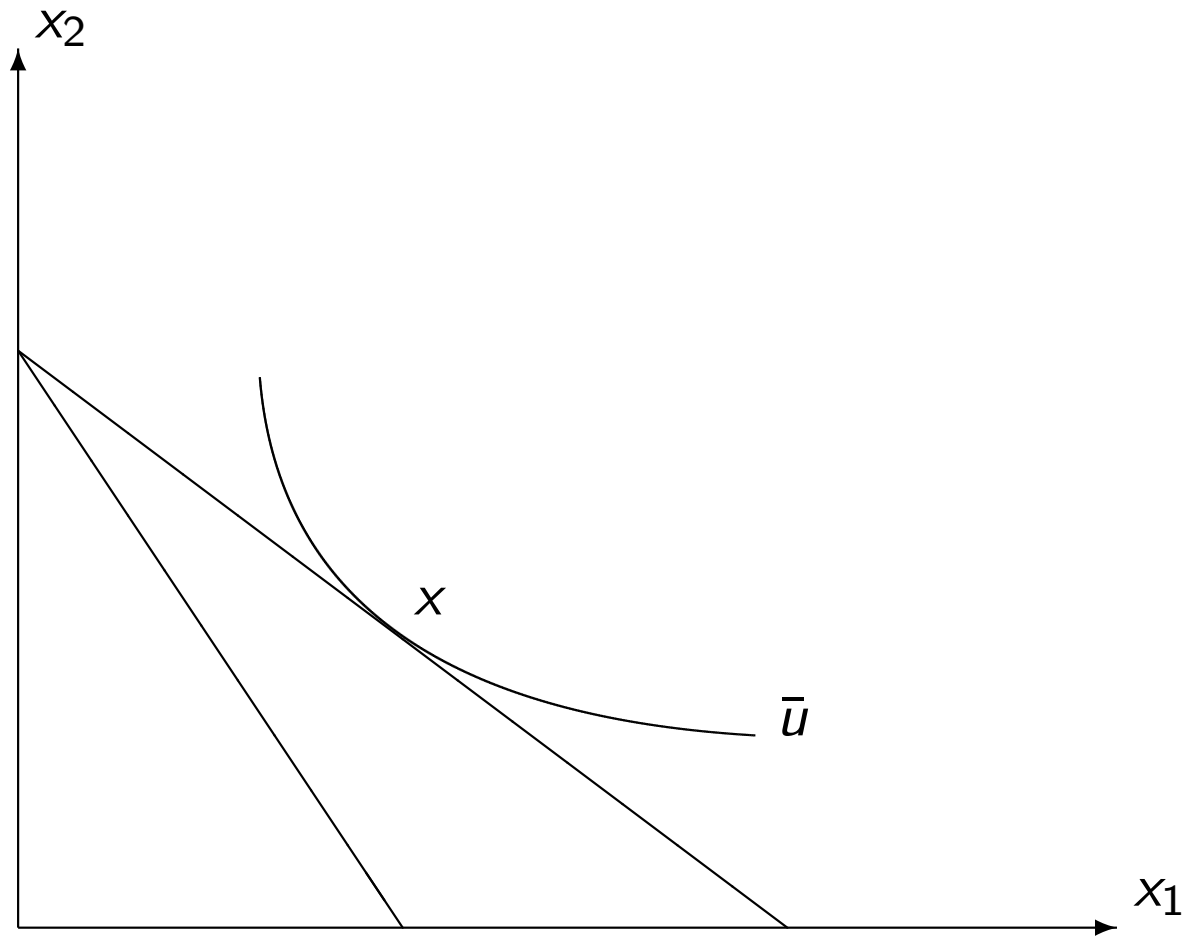
Definition (Equivalent variation)

The amount more wealth (i.e., the bonus) a consumer needs to achieve the same maximum utility at old prices (p) as she could achieve after a price change (to p'):

$$\text{EV} \equiv e\big(p, v(p',w)\big) - e\big(p', v(p',w)\big) = e\big(p, \underbrace{v(p',w)}_{\equiv \bar{u}'}\big) - w.$$

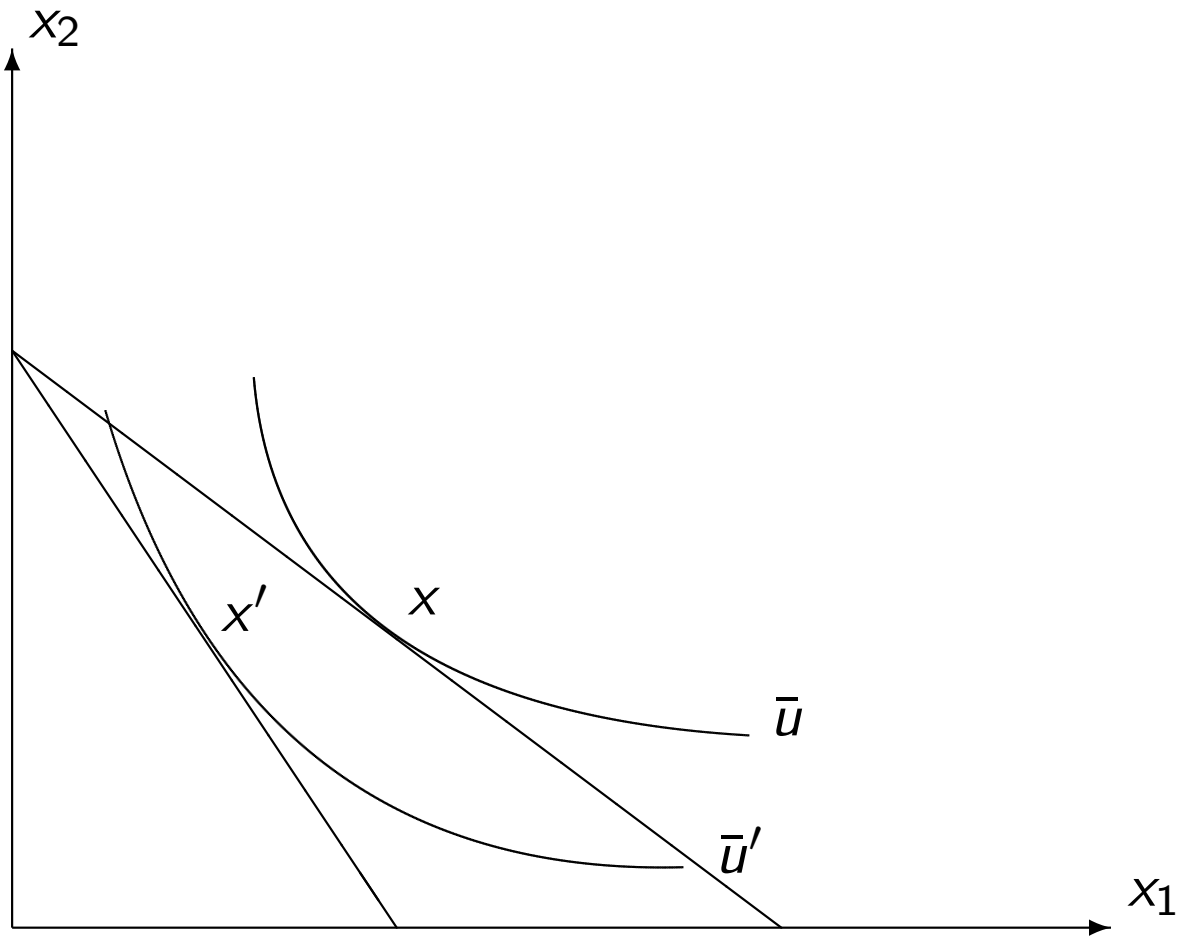
Illustrating compensating variation

- Suppose the price of good two is 1
- Price of good one increases



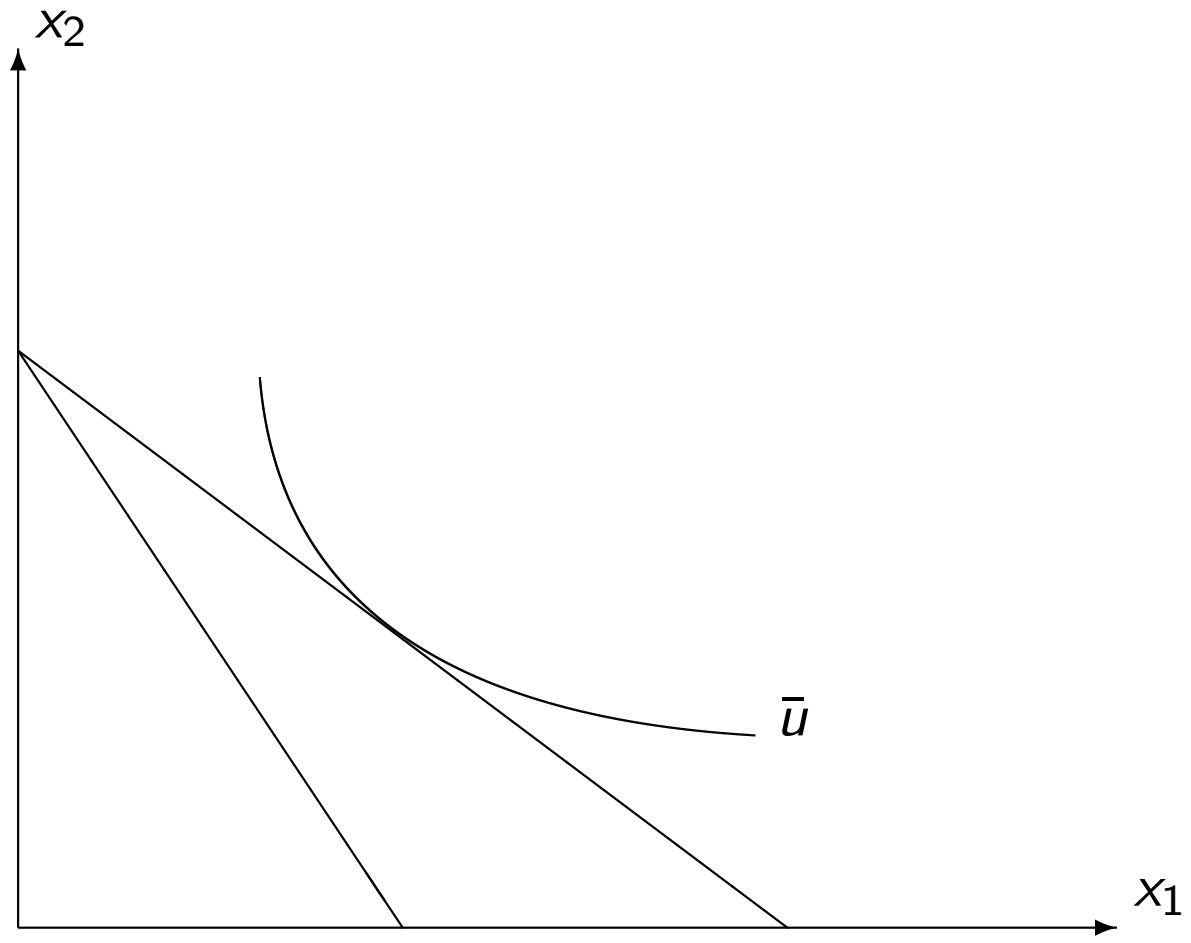
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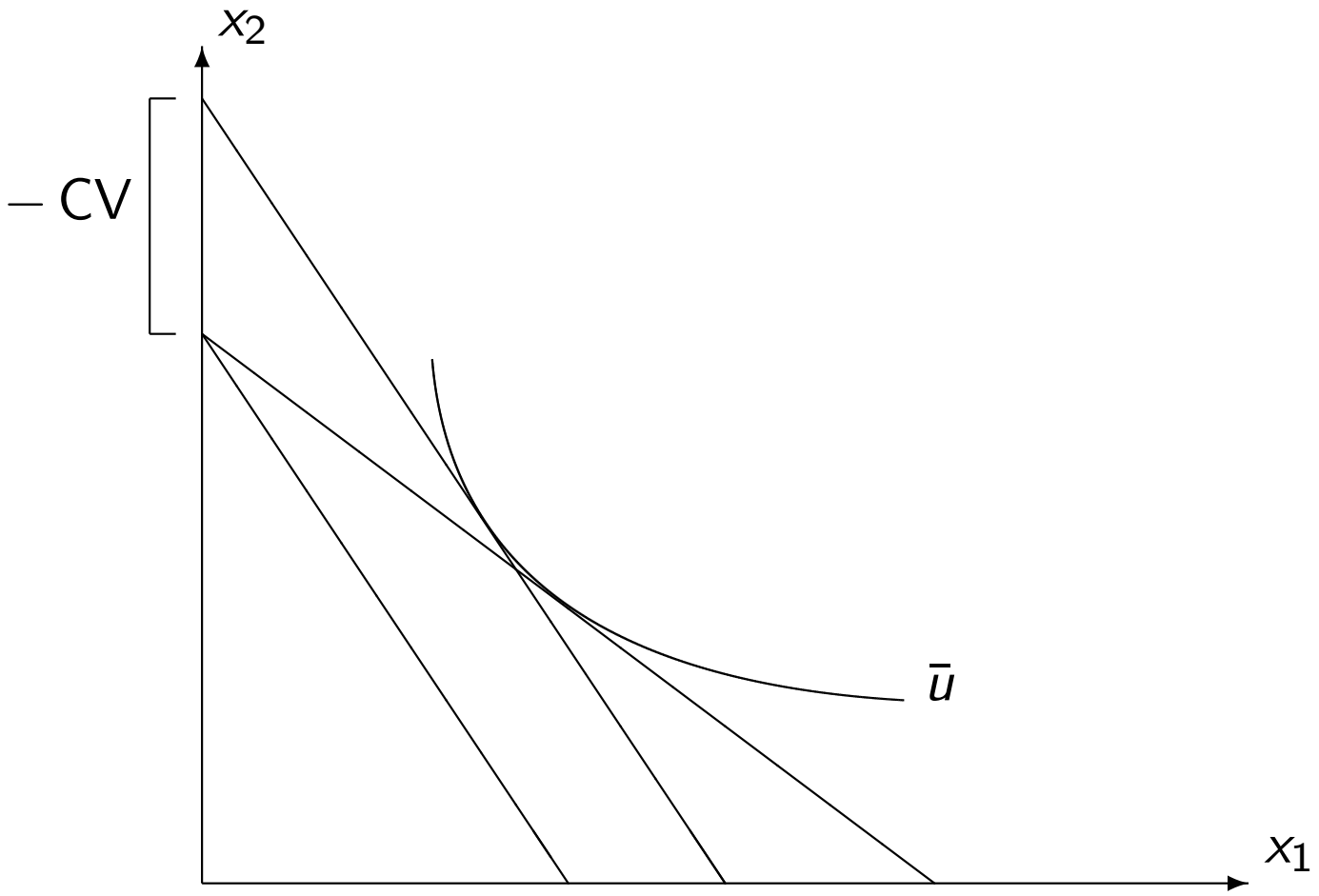
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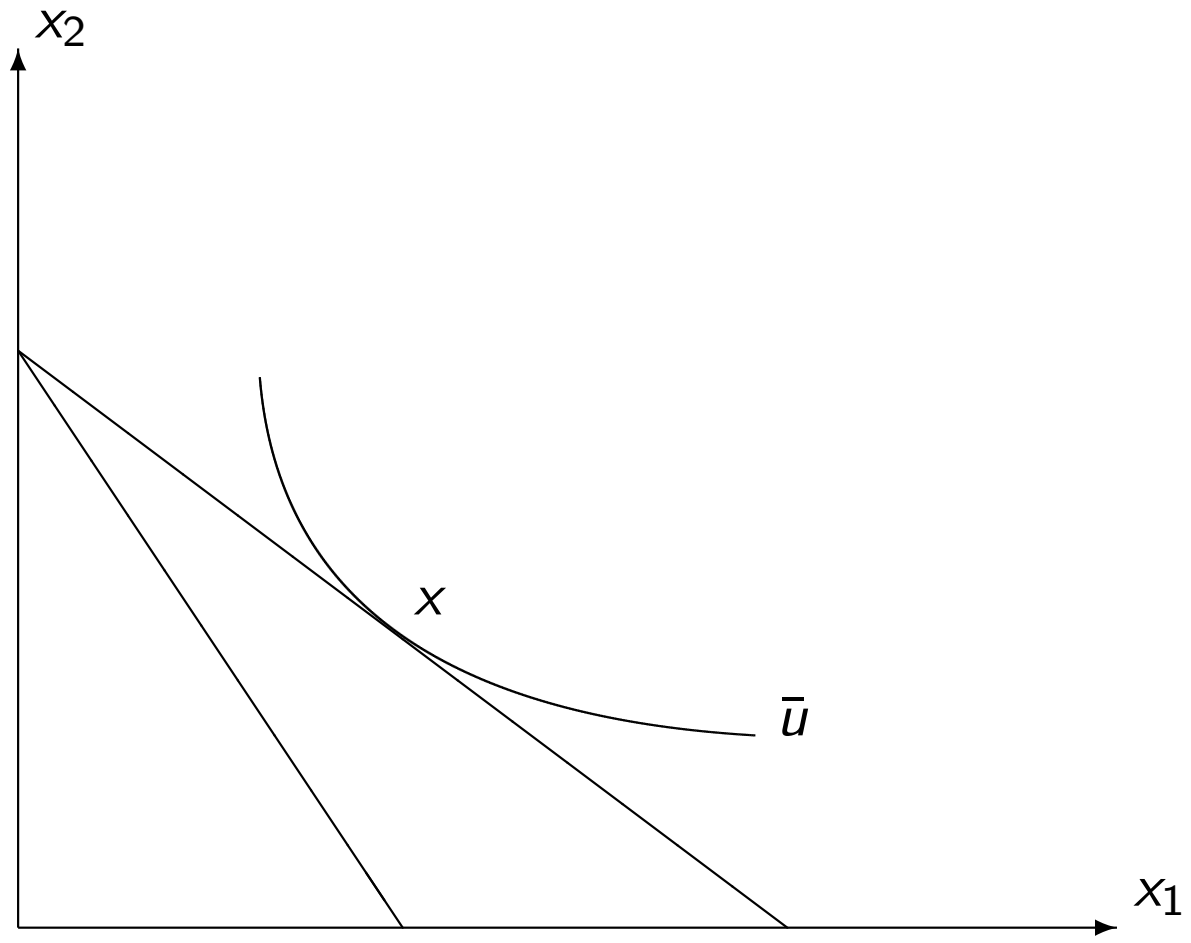
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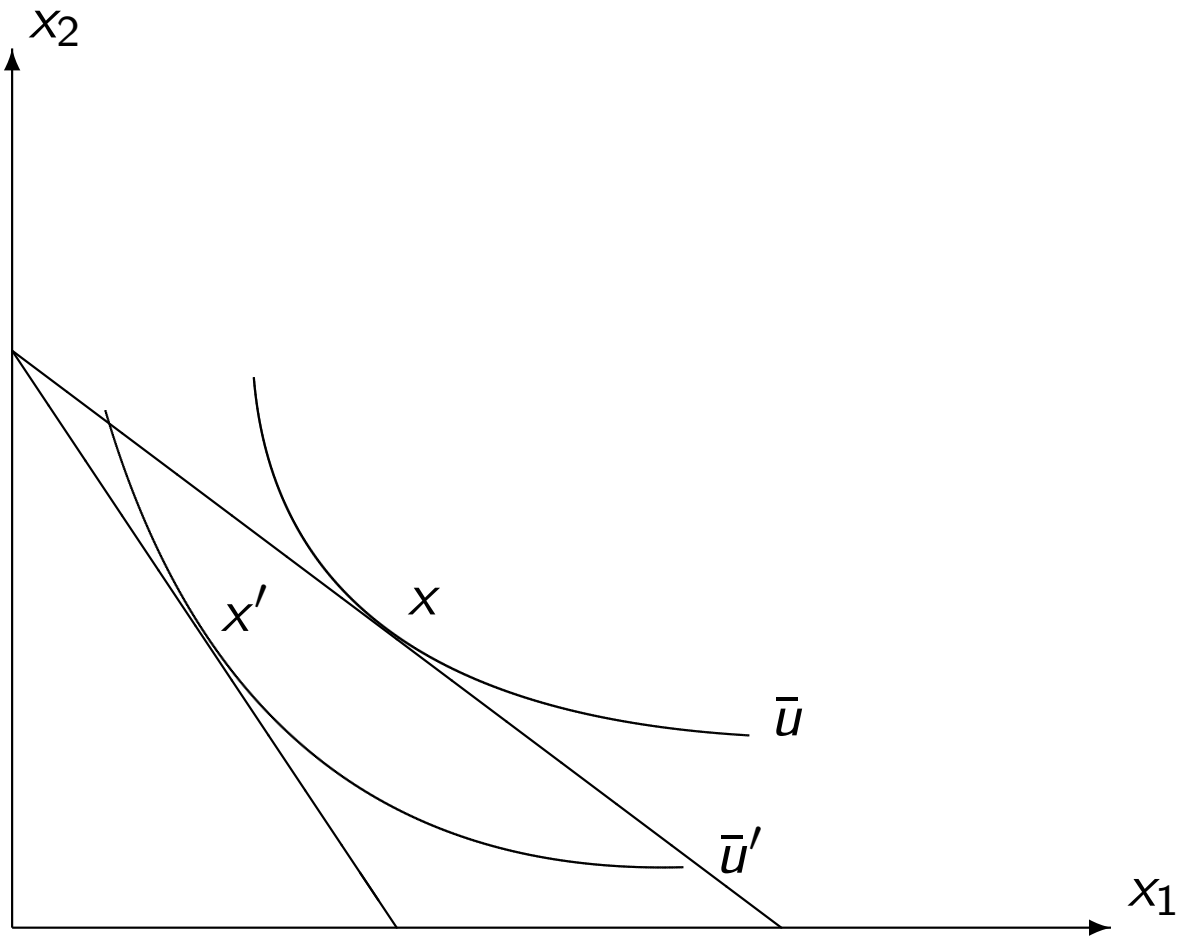
Illustrating equivalent variation

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- Price of good one increases



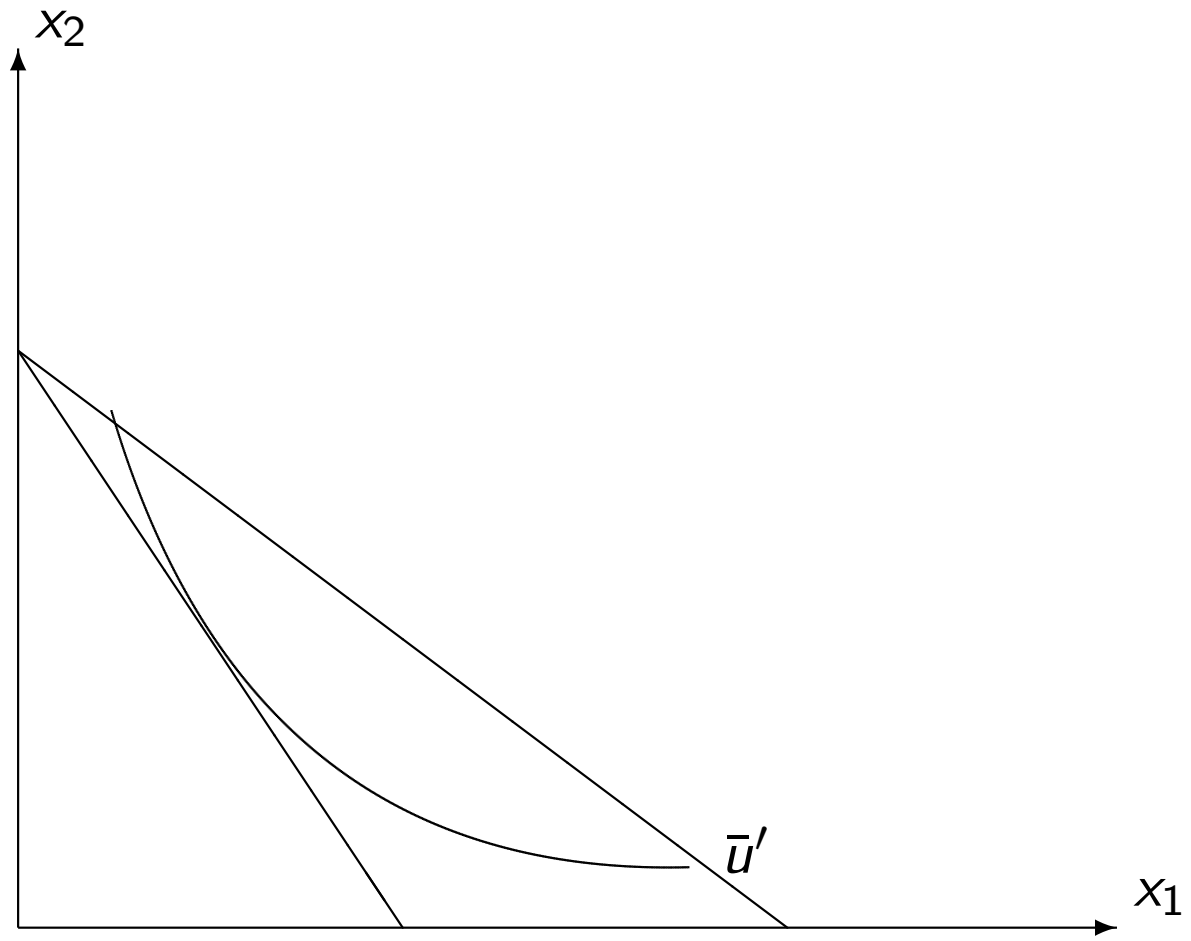
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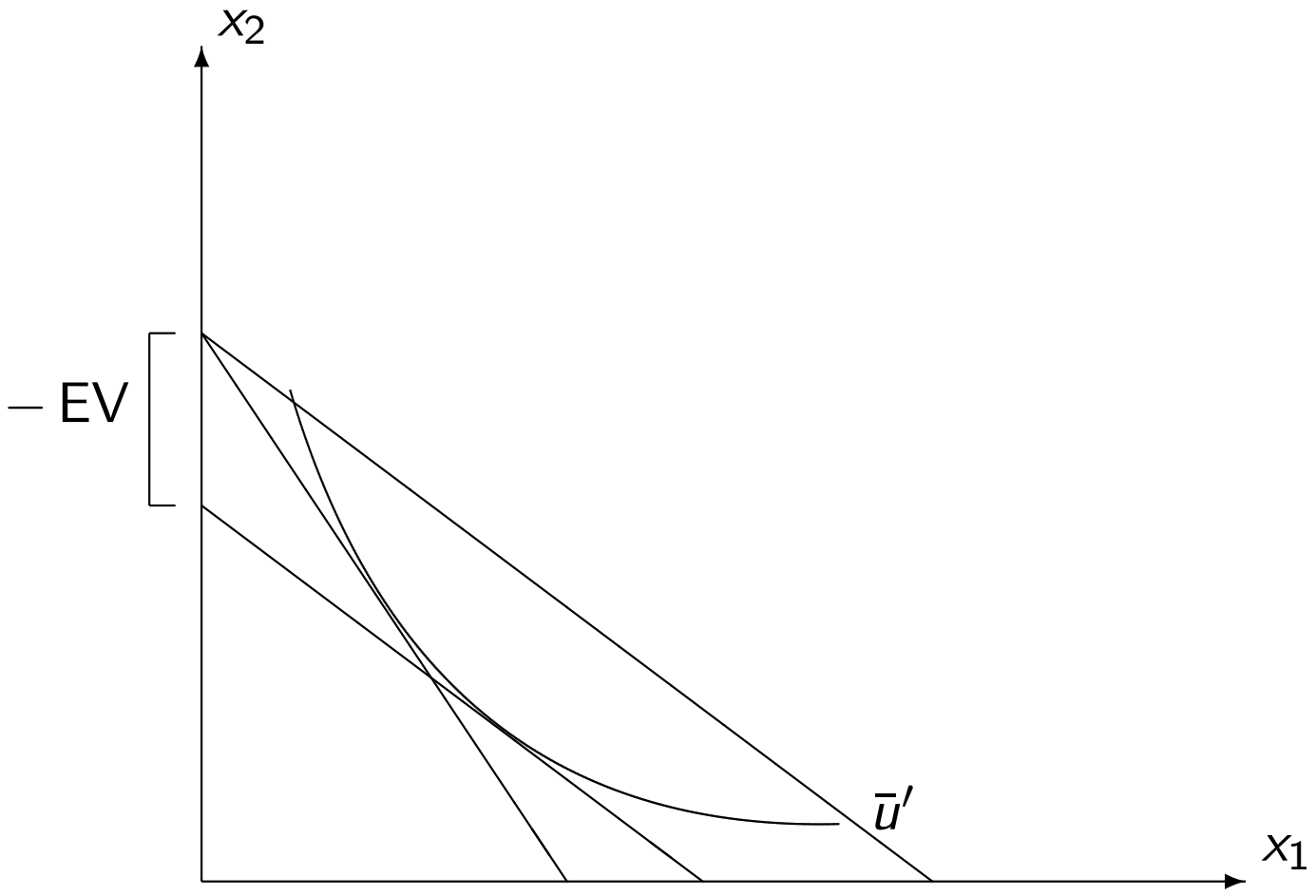
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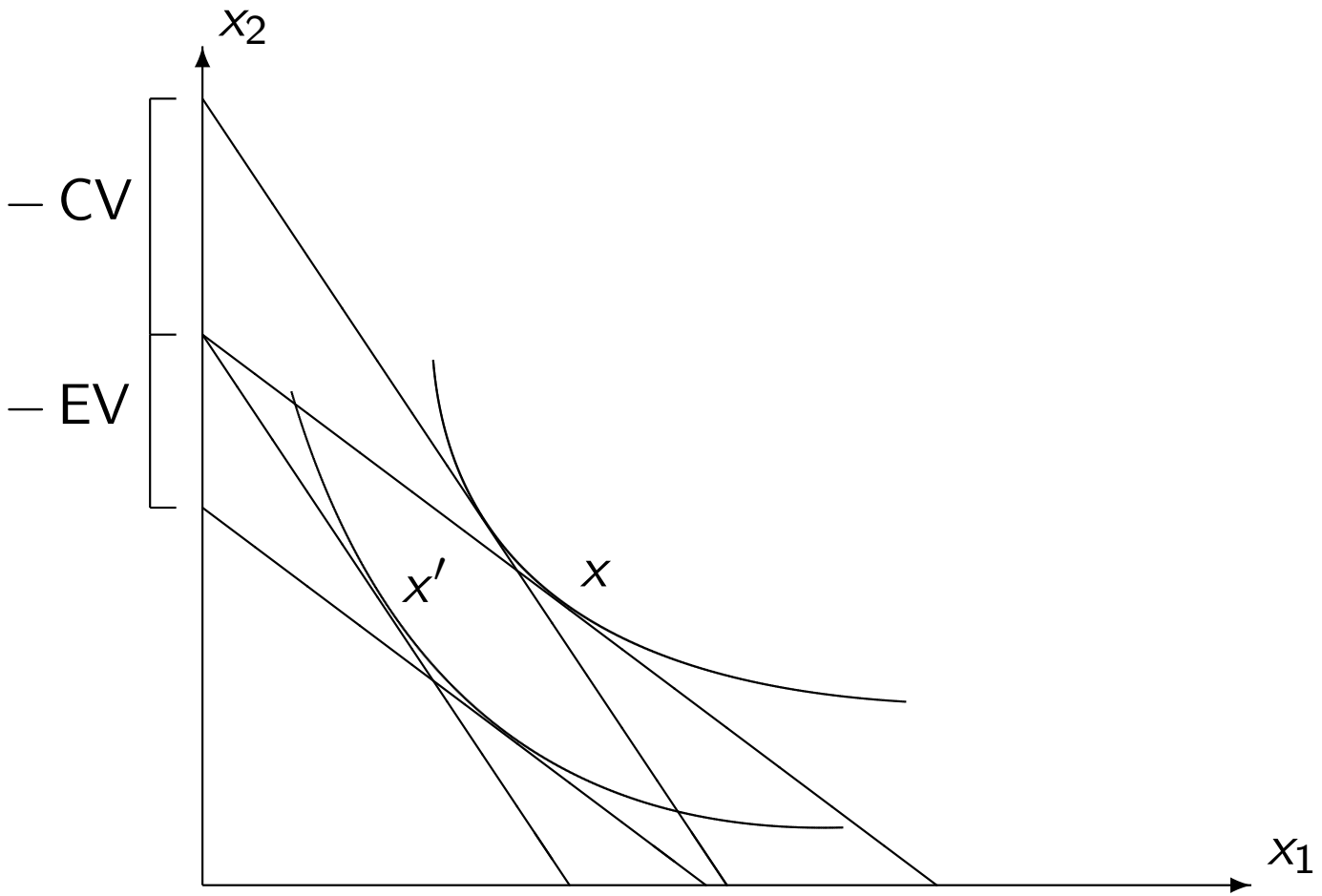
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We can't order CV and EV

- CV and EV are not necessarily equal
- We can't generally say which is bigger



Changing prices for a single good

Recall

$$\text{CV} = e(p, \bar{u}) - e(p', \bar{u})$$

Suppose the price of a single good changes from $p_i \longrightarrow p_i'$

$$= \int_{p_i'}^{p_i} \frac{\partial e(p, \bar{u})}{\partial p_i} \, dp_i$$

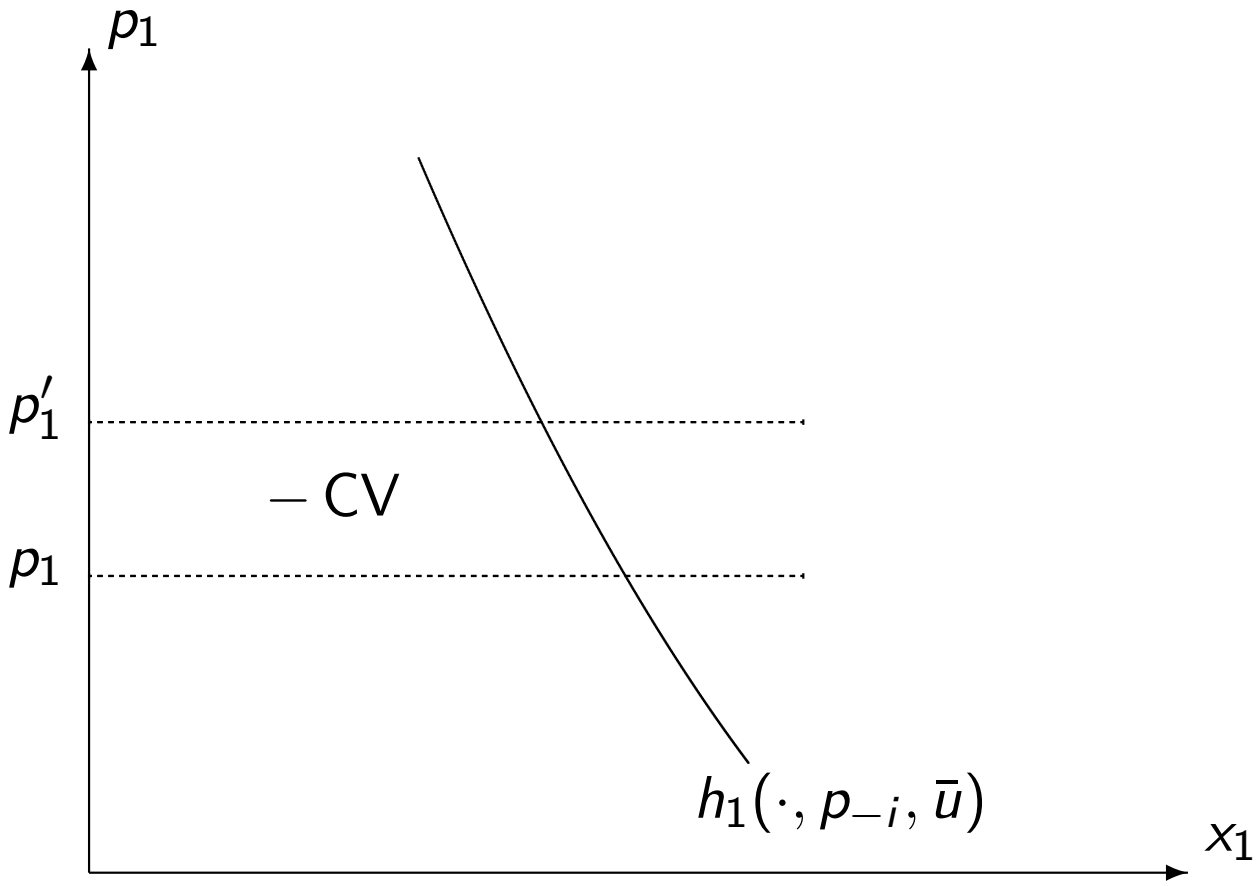
$$= \int_{p_i'}^{p_i} h_i(p, \bar{u}) \, dp_i = - \int_{p_i}^{p_i'} h_i(p, \bar{u}) \, dp_i$$

Similarly,

$$\text{EV} = \int_{p_i'}^{p_i} h_i(p, \bar{u}') \, dp_i = - \int_{p_i}^{p_i'} h_i(p, \bar{u}') \, dp_i$$

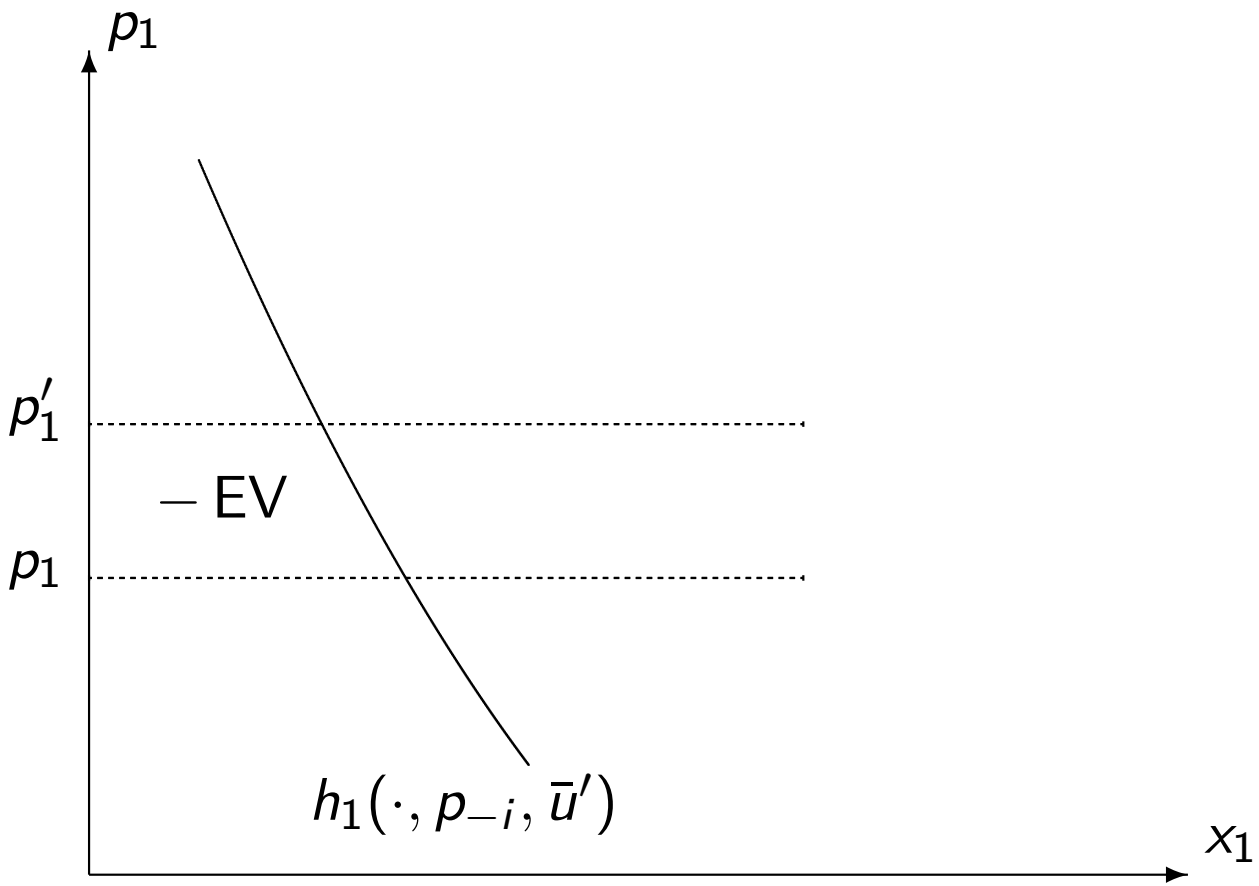
Illustrating changing prices for a single good: CV

- Suppose the price of good one increases from p_1 to p_1'
- Let $\bar{u} \equiv v(p, w)$ and $\bar{u}' \equiv v(p', w)$



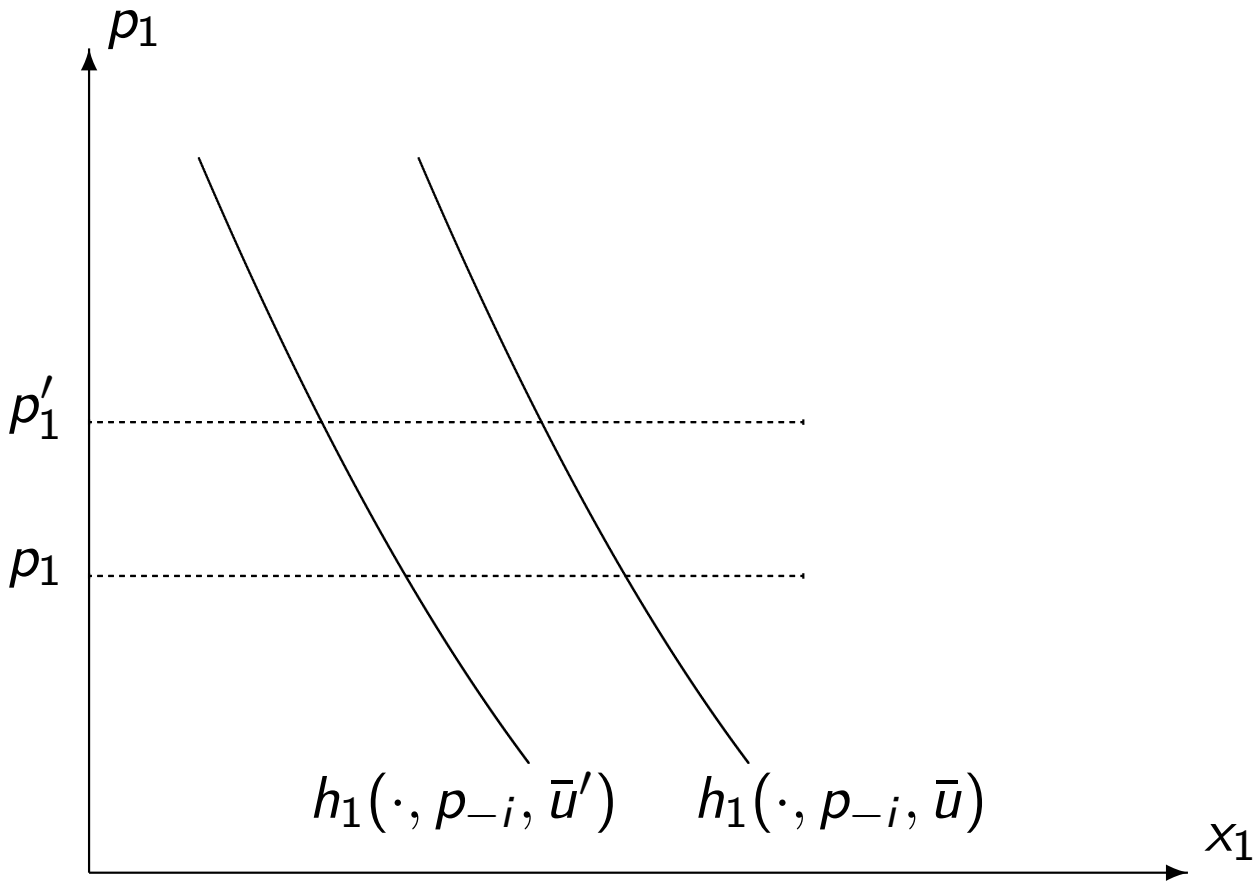
Illustrating changing prices for a single good: EV

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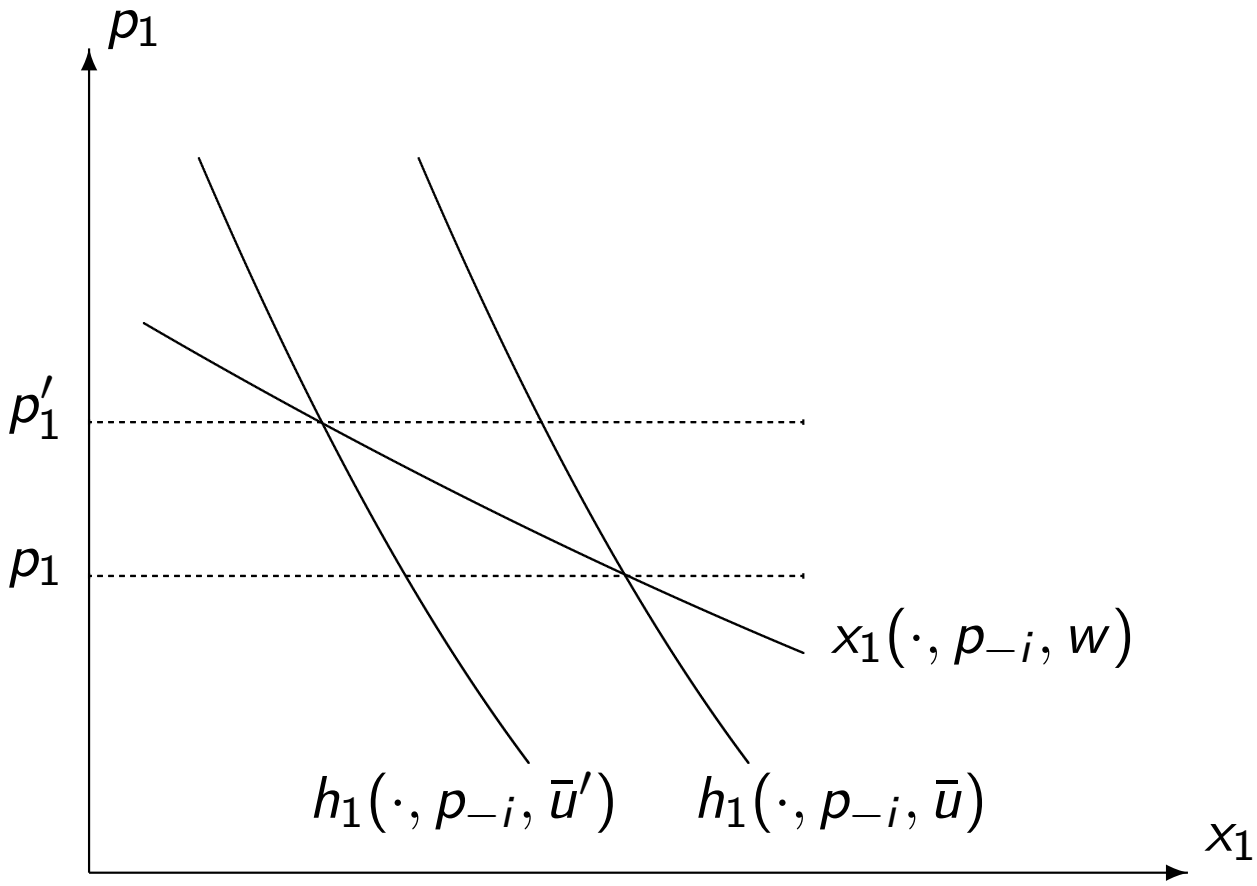
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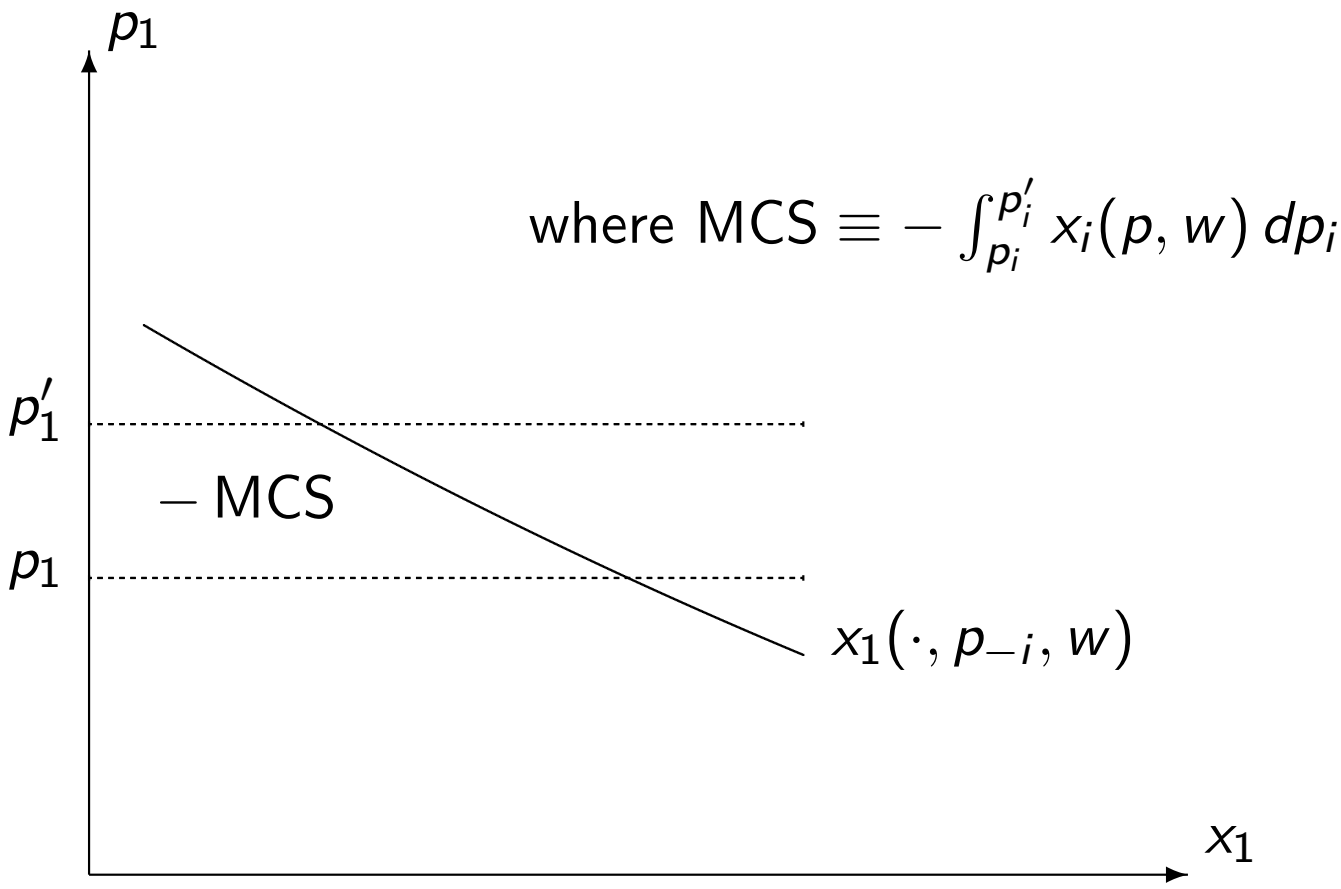
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Welfare and policy evaluation

- In theory, CV or EV can be summed across consumers to evaluate policy impacts
  - If $\sum_i \text{CV}_i > 0$, we can redistribute from "winners" to "losers," making everyone better off under the policy than before
  - If $\sum_i \text{EV}_i < 0$, we can redistribute from "losers" to "winners," making everyone better off than they would be if policy were implemented

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- In reality, identifying winners and losers is difficult
- In reality, widescale redistribution is generally impractical
- Sum-of-CV/EV criterion can cycle (i.e., it can look attractive to enact policy, and then look attractive to cancel it)

Outline

- The welfare impact of price changes
- Price indices
  - Price indices for all goods
  - Price indices for a subset of goods
- Aggregating across consumers
- Optimal taxation

Motivation for price indices

Problem: We generally can't access consumers' Hicksian demand correspondences (or even Marshallian ones)

We can say consumers are better off whenever wealth increases more than prices... but change of what prices?

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1. Ideally we would look at the changing "price" of a "util"
2. Since we can't measure utils, use change in weighted average of goods prices... but with what weights?

The Ideal index

The "price" of a "util" is expenditures divided by utility: $\frac{e(p,\bar{u})}{\bar{u}}$

Definition (ideal index)

$$\text{Ideal Index}(\bar{u}) \equiv \frac{p'_{\text{util}}}{p_{\text{util}}} = \frac{e(p',\bar{u})/\bar{u}}{e(p,\bar{u})/\bar{u}} = \frac{e(p',\bar{u})}{e(p,\bar{u})}.$$

Question: what $\bar{u}$ should we use?

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Question: what $\bar{u}$ should we use? Natural candidates are

- $v(p,w)$; note $e\big(p,v(p,w)\big) = w$, so denominator equals w
- $v(p',w')$; note $e\big(p',v(p',w')\big) = w'$, so numerator equals w'

Ideal index gives change in wealth required to keep utility constant

Weighted average price indices

We can't measure utility and don't know expenditure function $e(\cdot, \bar{u})$, so settle for an index based on weighted average prices

What weights should we use?

Weighted average price indices

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What weights should we use? Natural candidates are

- Quantity x of goods purchased at old prices p
- Quantity x' of goods purchased at new prices p'

The quantities used to calculated weighted average are often called the "basket"

Defining weighted average price indices

Definition (Laspeyres index)

$$\text{Laspeyres Index} \equiv \frac{p' \cdot x}{p \cdot x}$$

Definition (Paasche index)

$$\text{Paasche Index} \equiv \frac{p' \cdot x'}{p \cdot x'}$$

Defining weighted average price indices

Definition (Laspeyres index)

$$\text{Laspeyres Index} \equiv \frac{p' \cdot x}{p \cdot x} = \frac{p' \cdot x}{w} = \frac{p' \cdot x}{e(p, \bar{u})},$$

where $\bar{u} \equiv v(p, w)$.

Definition (Paasche index)

$$\text{Paasche Index} \equiv \frac{p' \cdot x'}{p \cdot x'} = \frac{w'}{p \cdot x'} = \frac{e(p', \bar{u}')}{p \cdot x'},$$

where $\bar{u}' \equiv v(p', w')$.

Bounding the Laspeyres and Paasche indices

Note that since $u(x) = \bar{u}$ and $u(x') = \bar{u}'$, by "revealed preference"

$$p' \cdot x \geq \min_{\xi:\, u(\xi) \geq \bar{u}} p' \cdot \xi = e(p', \bar{u})$$

$$p \cdot x' \geq \min_{\xi:\, u(\xi) \geq \bar{u}'} p \cdot \xi = e(p, \bar{u}')$$

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$$p \cdot x' \geq \min_{\xi:\ u(\xi) \geq \bar{u}'} p \cdot \xi = e(p, \bar{u}')$$

Thus we get that the Laspeyres index overestimates inflation, while the Paasche index underestimates it:

$$\text{Laspeyres} \equiv \frac{p' \cdot x}{e(p, \bar{u})} \geq \frac{e(p', \bar{u})}{e(p, \bar{u})} \equiv \text{Ideal}(\bar{u})$$

$$\text{Paasche Index} \equiv \frac{e(p', \bar{u}')}{p \cdot x'} \leq \frac{e(p', \bar{u}')}{e(p, \bar{u}')} \equiv \text{Ideal}(\bar{u}')$$

Why the Laspeyres and Paasche indices are not ideal

Deviation of Laspeyres/Paasche indices from Ideal comes from

$$p' \cdot x \geq p' \cdot h(p', \bar{u}) = e(p', \bar{u})$$
$$p \cdot x' \geq p \cdot h(p, \bar{u}') = e(p, \bar{u}')$$

The problem is that

- $p' \cdot x$ doesn't capture consumers' substitution away from x when prices change from p to p'
- $p \cdot x'$ doesn't capture consumers' substitution to x' when prices changed from p to p'

Particular forms of this substitution bias include

- New good bias
- Outlet bias

Price indices for a subset of goods

Suppose we can divide goods into two "groups"

1. Goods E: $\{1, \ldots, k\}$
2. Other goods $\{k+1, \ldots, n\}$

A meaningful price index for E requires that consumers can rank p_E without knowing p_{-E}

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A meaningful price index for E requires that consumers can rank p_E without knowing p_{-E}

For welfare ranking of price vectors for E not to depend on prices for other goods, we must have

$$e(p_E, p_{-E}, \bar{u}) \leq e(p'_E, p_{-E}, \bar{u}) \iff$$
$$e(p_E, p'_{-E}, \bar{u}') \leq e(p'_E, p'_{-E}, \bar{u}')$$

for all p_E, p'_E, p_{-E}, p'_{-E}, $\bar{u}$, and $\bar{u}'$

A "separability" result for prices

Recall

Theorem

Suppose $\succsim$ *on* $X \times Y$ *is represented by* $u(x, y)$*. Then preferences over* X *do not depend on* y *iff there exist functions* $v : X \to \mathbb{R}$ *and* $U : \mathbb{R} \times Y \to \mathbb{R}$ *such that*

1. $U(\cdot, \cdot)$ *is increasing in its first argument, and*
2. $u(x, y) = U\big(v(x), y\big)$ *for all* (x, y)*.*

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1. *$U(\cdot, \cdot)$ is increasing in its first argument, and*
2. *$u(x, y) = U\big(v(x), y\big)$ for all (x, y).*

Theorem

Welfare rankings over p_E do not depend on p_{-E} iff there exist functions $P : \mathbb{R}^k \to \mathbb{R}$ and $\hat{e} : \mathbb{R} \times \mathbb{R}^{n-k} \times \mathbb{R} \to \mathbb{R}$ such that

1. *$\hat{e}(\cdot, \cdot, \cdot)$ is increasing in its first argument, and*
2. *$e(p, \bar{u}) = \hat{e}\big(P(p_E), p_{-E}, \bar{u}\big)$ for all p and $\bar{u}$.*

Price indices for a subset of goods: other result

Results include that

- This separability in e gives that Hicksian demand for goods outside E only depend on p_E through the price index $P(p_E)$
- $P(\cdot)$ is homothetic (i.e., $P(p'_E) \geq P(p_E) \iff P(\lambda p'_E) \geq P(\lambda p_E)$); we can therefore come up with *some* $P(\cdot)$ which is homogeneous of degree one
- Neither of the two separability conditions defined by the theorems on the previous slide imply each other

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We can't model the individual consumers in an economy

There are typically too many consumers to model explicitly, so we consider a small number (often only one!)

- Valid if groups of consumers have same preferences and wealth
- If consumers are heterogeneous, validity of aggregation depends on
  - Type of analysis conducted
  - Form of heterogeneity

We consider several forms of analysis: under what forms of heterogeneity can we aggregate consumers?

Types of analysis conducted in the face of heterogeneity

We might try to

1. Model aggregate demand using only aggregate wealth
2. Model aggregate demand using wealth and preferences of a single consumer (i.e., a "positive representative consumer")
3. Model aggregate consumer welfare using welfare of a single consumer (i.e., a "normative representative consumer")

Modelling aggregate demand using aggregate wealth I

Question 1

Can we predict aggregate demand knowing only the aggregate wealth and not its distribution across consumers?

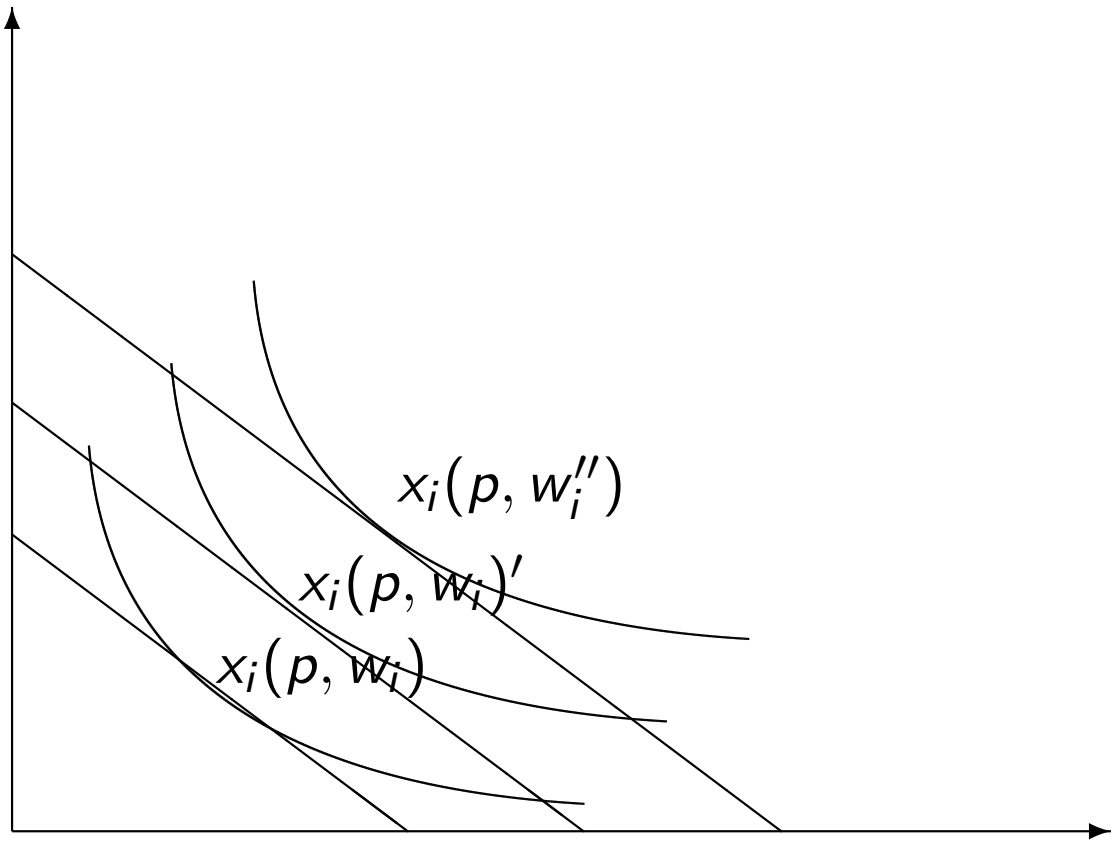
Necessary and sufficient condition: reallocation of wealth never changes total demand; i.e.,

$$\frac{\partial x_i(p, w_i)}{\partial w_i} = \frac{\partial x_j(p, w_j)}{\partial w_j}$$

for all p, i, j, w_i, and w_j

Modelling aggregate demand using aggregate wealth II

- Engle curves must be straight lines, parallel across consumers
- Consumers' indirect utility takes Gorman form:
  $v_i(p, w_i) = a_i(p) + b(p)w_i$



Aggregate demand with positive representative consumer

Question 2

Can aggregate demand be explained as though arising from utility maximization of a single consumer?

Aggregate demand with positive representative consumer

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Can aggregate demand be explained as though arising from utility maximization of a single consumer?

Answer: Not necessarily

Aggregate welfare with normative representative consumer

Question 3

Assuming there is a positive representative consumer, can her welfare be used as a proxy for some welfare aggregate of individual consumers?

Aggregate welfare with normative representative consumer

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Answer: Not necessarily

How does this work for firms?

Looking forward to our discussion of general equilibrium, we can also ask about aggregation across firms

Firms aggregate perfectly (assuming price-taking): given J firms,

- Aggregate supply as if single firm with production set

$$Y = Y_1 + \cdots + Y_J = \left\{ \sum_{j=1}^{J} y_j : y_j \in Y_j \text{ for each firm } j \right\}$$

- Profit function $\pi(p) = \sum_j \pi_j(p)$

Firms can aggregate because they have no wealth effects

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How should consumption be taxed I

Suppose we can impose taxes t in order to fund some spending T
What taxes should we impose? Several ways to approach this

1. Maximize $v(p+t, w)$ such that $t \cdot x(p+t, w) \geq T$
2. Minimize $e(p+t, \bar{u})$ such that $t \cdot h(p+t, \bar{u}) \geq T$

Following the second approach gives Lagrangian

$$\mathcal{L} = -e(p+t, \bar{u}) + \lambda\big(t \cdot h(p+t, \bar{u}) - T\big)$$

And FOC

$$\nabla_p e(p+t^*, \bar{u}) = \lambda h(p+t^*, \bar{u}) + \lambda\big[\nabla_p h(p+t^*, \bar{u})\big]\, t^*$$

How should consumption be taxed II

$$\underbrace{\nabla_p e(p + t^*, \bar{u})}_{h(p+t^*,\bar{u})} - \lambda h(p + t^*, \bar{u}) = \lambda \left[\nabla_p h(p + t^*, \bar{u})\right] t$$

$$\frac{1 - \lambda}{\lambda} h(p + t^*, \bar{u}) = \left[\nabla_p h(p + t^*, \bar{u})\right] t^*$$

$$\frac{1 - \lambda}{\lambda} \left[\nabla_p h(p + t^*, \bar{u})\right]^{-1} h(p + t^*, \bar{u}) = t^*$$

This is a generally a difficult system to solve

The no-cross-elasticity case

If $\frac{\partial h_i}{\partial p_j} = 0$ for $i \neq j$, we can solve on a tax-by-tax basis:

$$\lambda t_i^* \frac{\partial h_i(p + t^*, \bar{u})}{\partial p_i} = \underbrace{\frac{\partial e(p + t^*, \bar{u})}{\partial p_i}}_{= h_i(p + t^*, \bar{u})} - \lambda h_i(p + t^*, \bar{u})$$

$$\lambda t_i^* \frac{\partial h_i(p + t^*, \bar{u})}{\partial p_i} = (1 - \lambda) h_i(p + t^*, \bar{u})$$

$$t_i^* = \frac{1 - \lambda}{\lambda} h_i(p + t^*, \bar{u}) \left[\frac{\partial h_i(p + t^*, \bar{u})}{\partial p_i} \right]^{-1}$$

$$\frac{t_i^*}{p_i} = \frac{1 - \lambda}{\lambda} \left[\frac{\partial h_i(p + t^*, \bar{u})}{\partial p_i} \frac{p_i}{h_i(p + t^*, \bar{u})} \right]^{-1}$$

So optimal tax rates are proportional to the inverse of the elasticity of Hicksian demand